

5	(a) (i)	ratio of charge and potential (difference)/ voltage (ratio must be clear)	B1	[1]
	(ii)	capacitor has equal magnitudes of (+)ve and (-)ve charge <u>total</u> charge on capacitor is zero (so does not store charge) (+)ve and (-)ve charges to be separated work done to achieve this so stores energy	B1 B1 M1 A1	[4]
	(b) (i)	capacitance of Y and Z together is $24 \mu\text{F}$ $1/C = 1/24 + 1/12$ $C = 8.0 \mu\text{F}$ (allow 1 s.f.)	C1 A1	[2]
	(ii)	some discussion as to why all charge of one sign on one plate of X $Q = (CV =) \underline{8.0 \times 10^{-6}} \times 9.0$ $= 72 \mu\text{C}$	B1 M1 A0	[2]
	(iii) 1.	$V = (72 \times 10^{-6}) / (12 \times 10^{-6})$ $= 6.0 \text{ V}$ (allow 1 s.f.) (allow 72/12)	A1	[1]
	2.	either $Q = 12 \times 10^{-6} \times 3.0$ or charge is shared between Y and Z charge = $36 \mu\text{C}$ Must have correct voltage in (iii)1 if just quote of $36 \mu\text{C}$ in (iii)2.	C1 A1	[2]

6	(a) (i)	particle must be moving	M1	
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